

RIS-Assisted Jamming Rejection and Path Planning for UAV-Borne IoT Platform: A New Deep Reinforcement Learning Framework

Shuyan Hu¹, Member, IEEE, Xin Yuan², Member, IEEE, Wei Ni², Senior Member, IEEE, Xin Wang¹, Fellow, IEEE, and Abbas Jamalipour³, Fellow, IEEE

Abstract—This article presents a new deep reinforcement learning (DRL)-based approach to the trajectory planning and jamming rejection of an unmanned aerial vehicle (UAV) for the Internet of Things (IoT) applications. Jamming can prevent timely delivery of sensing data and reception of operation instructions. With the assistance of a reconfigurable intelligent surface (RIS), we propose to augment the radio environment, suppress jamming signals, and enhance the desired signals. The UAV is designed to learn its trajectory and the RIS configuration based solely on changes in its received data rate, using the latest deep deterministic policy gradient (DDPG) and twin delayed DDPG (TD3) models. Simulations show that the proposed DRL algorithms give the UAV with strong resistance against jamming and that the TD3 algorithm exhibits faster and smoother convergence than the DDPG algorithm, and suits better for larger RISs. This DRL-based approach eliminates the need for knowledge of the channels involving the RIS and jammer, thereby offering significant practical value.

Index Terms—Deep deterministic policy gradient (DDPG), Internet of Things (IoT), jamming rejection, reconfigurable intelligent surface (RIS), twin delayed DDPG (TD3), unmanned aerial vehicle (UAV).

I. INTRODUCTION

THE USE of unmanned aerial vehicles (UAVs) has been growing in popularity with the expansion of the Internet of Things (IoT) [1], [2], [3]. For instance, UAVs equipped with sensors and cameras are increasingly deployed to monitor and collect data on air quality, traffic, and other environmental factors for IoT applications [4]. The data collected by the UAVs need to be transmitted back to a control center or ground

base station (BS) for data analysis or for triggering automated responses, such as adjusting traffic lights to alleviate congestion [5].

Jamming can have severe effects on the physical-layer security of UAV-borne IoT applications [6], [7], [8], [9]. An attacker can use jamming equipment to disrupt the wireless transmissions between the UAVs and the BS, preventing the UAVs from transmitting their sensing data and receiving operation instructions in a timely manner and causing the UAVs to fail their missions [10]. As a matter of fact, jamming has been identified to be one of the most critical threats to the massive IoT connections in the context of the upcoming sixth-generation (6G) communication systems [11].

Identifying and eliminating jammers is challenging without specialized equipment, such as radar [12]. Previous research on jamming cancelation for UAV-borne IoT applications has not specifically focused on UAV-borne IoT platforms. Duo et al. [13] targeted to improve the worst case long-term data rate for a UAV-assisted wireless sensor network under jamming attacks by jointly configuring the UAV's transmission strategy and 3-D flight path. In [14], a reinforcement learning (RL)-based communication strategy was developed between a UAV swarm and a BS to counteract jamming attempts. This improved the communication quality of UAVs by exploring the motion and antenna spatial domain. Li et al. [15] proposed a secure UAV communication scheme against smart jammers using a knowledge-based RL approach, which leveraged domain information to reduce the state space and speed up the convergence of the RL algorithm.

On the other hand, reconfigurable intelligent surfaces (RISs) were developed as a means of creating programmable wireless transmission environments [16]. These surfaces, made up of passive reflecting units with reprogrammable phase shifts [17], can be placed on building surfaces, and are expected to be an integral part and effective enhancer for the IoT. By configuring the phase shifts, the RIS-reflected signals are added constructively to the direct signal to improve signal quality or destructively to reduce interference [18]. The performance gain of RIS-assisted wireless communication was experimentally validated in [19].

The potential of RISs for anti-jamming applications has not been widely studied [20], [21]. In [22], a collective active and passive beamforming design was formulated to lower transmit power by optimizing the continuous phase shifts of an RIS.

Manuscript received 6 February 2023; revised 1 May 2023; accepted 3 June 2023. Date of publication 7 June 2023; date of current version 7 November 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62231010, Grant 62071126, and Grant 62101135; and in part by the Innovation Program of Shanghai Municipal Science and Technology Commission under Grant 20JC1416400 and Grant 21XD1400300. (Shuyan Hu and Xin Yuan contributed equally to this work.) (Corresponding author: Xin Wang.)

Shuyan Hu and Xin Wang are with the Key Laboratory of EMW Information (MoE), Department of Communication Science and Engineering, Fudan University, Shanghai 200433, China (e-mail: syhu14@fudan.edu.cn; xwang11@fudan.edu.cn).

Xin Yuan and Wei Ni are with the Data61, Commonwealth Scientific and Industrial Research Organization, Sydney, NSW 2122, Australia (e-mail: xin.yuan@data61.csiro.au; wei.ni@data61.csiro.au).

Abbas Jamalipour is with the School of Electrical and Information Engineering, The University of Sydney, Sydney, NSW 2006, Australia (e-mail: a.jamalipour@ieee.org).

Digital Object Identifier 10.1109/JIOT.2023.3283502

TABLE I
DIFFERENCE BETWEEN THIS WORK AND EXISTING WORKS

Topic Work	Anti-jamming UAV comm.	RIS-assisted anti-jamming comm.	RIS-assisted UAV comm.
[14]–[16]	✓		
[22]–[26]		✓	
[27]–[36]			✓
This paper	✓	✓	✓

The result was later extended to discrete phase shifts in [23]. The RIS-assisted secure transmission was studied in [24], [25], and [26], where fast RL and deep RL (DRL) were used to design the active and passive beamformers. For instance, in [24], jamming signals were modulated and reflected by an RIS to reduce the eavesdropping data rate.

In the context of UAV-borne IoT platforms, RIS-assisted jamming rejection has yet to be addressed. The design of an RIS-assisted jamming rejection UAV-borne IoT platform faces the following challenges. First, the received data rate at the UAV is a nonconvex function of the UAV's flight path and the RIS phase shifts. Second, the UAV flight path waypoints are embedded in the exponents of the communication link, making the trajectory optimization intractable for existing convex tools, such as successive convex approximation. In addition, the large number of RIS reflecting units can cause prohibitive overhead and complexity for radio channel estimation, acquisition, and reconfiguration.

In a different yet relevant context, some studies have examined general RIS-assisted UAV communications [27], [28], [29], [30]. Li et al. [27] jointly optimized UAV flight path and RIS passive beamformers to achieve the largest average rate of terrestrial users. In [29], a UAV and RIS were configured to deliver ultrareliable and low-latency commands among terrestrial IoT devices using nonconvex optimization. UAVs and RISs have been considered to assist with communications in a vehicular network [31], IoT [32], and wireless powered communication network [33]. For instance, the coverage probability, bit error rate, block error rate, and throughput of a vehicular network were analyzed in [31]. Alternating optimization was employed in [33] to maximize the sum rate of energy harvesting-powered IoT nodes. Moreover, UAV-mounted RISs have been also employed to assist communication or data collection in a multiuser network [34], [35], [36]. The sum rate of nonorthogonal multiple-access was maximized in [34] by block coordinate descent and Riemannian conjugate gradient. The average throughput was analyzed in [35]. The number of UAVs was minimized by optimizing the RIS deployment and assignment by integer linear programming in [36]. A summary of existing works is provided in Table I.

The practicality and advantages of RIS-assisted UAV communications were experimentally corroborated by computer simulations in [27], [28], [29], [30], [34], [35], and [36]. A UAV-borne RIS was designed for communication enhancement using experimental results from UAV hovering in [37]. On the other hand, an RIS is typically controlled via a field programmable gate array (FPGA) [38]. The execution delays

of FPGA units are in nanoseconds [39]. Therefore, FPGA-controlled RISs can achieve a phase shift change with a delay of microseconds. Consider a UAV's typical speed of about 30 m/s. The reconfiguration of an RIS is fast enough to cater to the time-varying UAV communication channels. It is feasible to introduce RIS into UAV communications.

In this article, we put forth a new DRL-based architecture for the flight path planning and jamming cancelation of UAV-borne IoT platforms. A fixed-wing UAV is considered. Our architecture utilizes an RIS, which dynamically modifies the wireless transmission environment to mitigate jamming power and enhance intended signals to the UAV. To accomplish this, we devise a new deep deterministic policy gradient (DDPG) model and its enhancement, Twin delayed DDPG (TD3), to allow the UAV to learn its flight path and the RIS configuration based solely on its received data rate, eliminating the need for the channel state information (CSI) involving the RIS and jammer in the flight path training. This presents a significant practical advantage, as the estimation of CSI involving an RIS is complex and may not be able to be performed in real-time.

The main contributions of this article are as follows.

- 1) A new problem is introduced to jointly optimize flight path planning and RIS-assisted jamming cancelation to maximize the data rate of a UAV-borne IoT platform. The problem is nonstraightforward for its nonconvexity and sequential decision-making nature.
- 2) A new DRL architecture is proposed to solve the new problem and allow the UAV to learn its flight path and the RIS configuration based solely on its received data rate, eliminating the need for CSI in the training process.
- 3) The DRL architecture is implemented using the latest DRL models, DDPG, and its twin-delayed version, i.e., TD3. While TD3 is generally applicable to the problem under investigation, DDPG can benefit from its simpler network architecture and smooth convergence and is suitable for problems with smaller scales, e.g., fewer RISs, and less stringent mission time requirements.

The proposed DRL approach is validated through extensive simulations, showing its exceptional resistance against jamming. Particularly, the DDPG demonstrates faster and smoother convergence when the mission time is long and the RIS is small. The TD3 outperforms the DDPG in robustness against the jammer's position, especially when the mission time is short. This is crucial, as accurately locating the jammer and estimating its CSI can be practically challenging.

The remainder of this article is arranged as follows. In Section II, the system model is described. In Section III, we formulate the problem of jointly designing the UAV's flight path and the RIS configuration to maximize the data rate in the presence of an unknown jammer. In Section IV, we propose new DRL solutions to the problem. Performances are gauged in Section V. This article is concluded in Section VI.

Notation: Boldface lower and upper cases indicate vectors and matrices, respectively; $\mathbb{C}^N$ denotes the space of $N \times 1$ complex-valued column vectors; $\|\cdot\|$ denotes the Euclidean norm; $(\cdot)^T$ and $(\cdot)^H$ stand for transpose and conjugate transpose, respectively; $\text{diag}(\mathbf{a})$ is a diagonal matrix with the

TABLE II
LIST OF NOTATIONS AND VARIABLES

Notation	Description
T, T_w	Scheduling period and total number of time slots
t, δ	Index and duration of each time slot
N	Total number of RIS reflecting elements
N_x, N_y	Number of RIS elements on the x - and y -axis
$\mathbf{q}_B, \mathbf{q}_J, \mathbf{q}_R$	3D locations of the BS, Jammer, and RIS
$\mathbf{q}_0, \mathbf{q}_F$	Initial and final 3D waypoints of the UAV
$\mathbf{q}_t, \mathbf{V}_t, \mathbf{A}_t$	3D waypoint, velocity and acceleration of the UAV at time slot t
ϑ	Maximum pitch angle of the UAV
θ_n^t, Θ_t	RIS phase shift of each element and phase shift matrix at time slot t
$d_{BU}^t, d_{JU}^t, d_{RU}^t$	Distance from the UAV to the BS, Jammer, and RIS at time slot t
d_{BR}, d_{JR}	Distance from the RIS to the BS and Jammer
β_t, β	Rician factors
κ_1, κ_2	Path loss exponents
$h_{BU}^t, h_{JU}^t, h_{RU}^t$	Channel gains of the UAV-BS, UAV-Jammer, and UAV-RIS links at time slot t
h_{BR}, h_{JR}	Channel gains of the BS-RIS and Jammer-RIS links
P_t, P_J	Transmit power of the BS and Jammer
γ_U^t, R_U^t	Signal-to-interference-plus-noise ratio (SINR) and data rate of the UAV at time slot t
s_t, a_t, r_t	State, action, and reward at time slot t
θ_a, θ_c	Parameters of the training-actor and training-critic networks
θ'_a, θ'_c	Parameters of the target-actor and target-critic networks

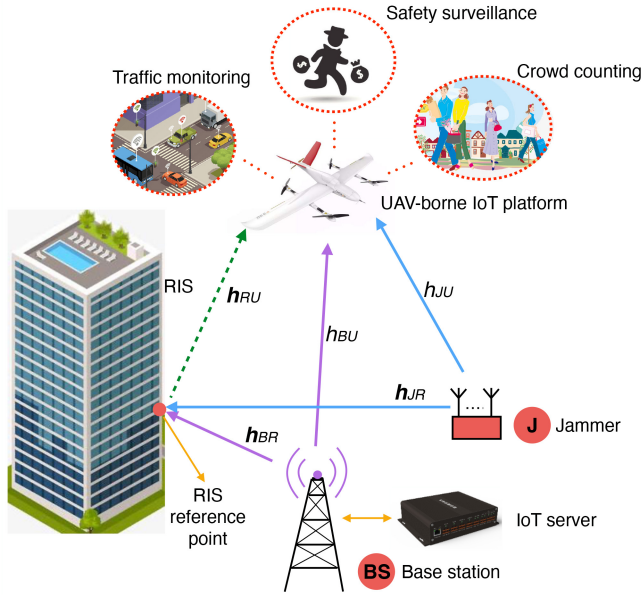


Fig. 1. RIS-assisted anti-jamming UAV-borne IoT platform, where an RIS is adaptively configured along with the UAV trajectory to enhance the desired signals and reject the jamming signals.

elements of $\mathbf{a}$ along the diagonal; $\otimes$ stands for the Kronecker product. Other notations are listed in Table II.

II. SYSTEM MODEL

As depicted in Fig. 1, a ground BS offers wireless communication service for a fixed-wing UAV in the presence of a terrestrial jammer. Consider a 3-D Cartesian coordinate system. The BS and the jammer are placed at $\mathbf{q}_B = [0, 0, 0]^T$ and $\mathbf{q}_J = [x_J, y_J, 0]^T$. An RIS is deployed to facilitate the UAV communication and suppress the jamming signals from

the jammer. We suppose that the BS, UAV, and jammer all have a single antenna for description convenience (but the proposed DRL architecture can be extended to support multi-antenna BSs and UAVs, as part of our future work). Only based on its received data rate, the UAV determines its flight path and the RIS configuration dynamically. The RIS is installed with a smart controller to procure the command from the UAV (via the cellular system) for RIS configuration [40].

A. UAV Mobility Model

The UAV functions for a finite scheduling horizon of T seconds, which is split into T_w time slots indexed by t , and $t = 1, \dots, T_w$. A slot lasts $\delta = T/T_w$, which is short enough that the UAV can be viewed as stationary per slot. The UAV is traveling from a predetermined starting position $\mathbf{q}_0 = [x_0, y_0, z_0]^T$ to a predetermined ending position $\mathbf{q}_F = [x_F, y_F, z_F]^T$. The UAV's 3-D coordinates are $\mathbf{q}_t = [x_t, y_t, z_t]^T \forall t$. Let $\mathbf{V}_t := [V_{xt}, V_{yt}, V_{zt}]^T$ and $\mathbf{A}_t := [A_{xt}, A_{yt}, A_{zt}]^T$ collect the velocity and acceleration of the UAV per slot t , respectively. The UAV obeys some mobility constraints [41]

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \mathbf{V}_t \delta + \frac{1}{2} \mathbf{A}_t \delta^2 \quad \forall t \quad (1a)$$

$$\mathbf{q}_F = \mathbf{q}_{T_w} \quad (1b)$$

$$\mathbf{V}_{t+1} = \mathbf{V}_t + \mathbf{A}_t \delta \quad \forall t \quad (1c)$$

$$|A_{i,t}| \leq A_{\max}, i \in \{x, y, z\} \quad \forall t \quad (1d)$$

$$V_{\min} \leq \|\mathbf{V}_t\| \leq V_{\max} \quad \forall t \quad (1e)$$

$$V_{zt}/\|\mathbf{V}_t\| \leq \sin \vartheta \quad \forall t \quad (1f)$$

where $A_{\max}$ is the UAV's largest acceleration; $V_{\max}$ and $V_{\min}$ are the UAV's maximum and minimum speeds; and ϑ is the largest UAV pitch angle when ascending or descending.

B. RIS Configuration

The RIS is adhered on the outer surface of a building, which is on the (x, z) -plane and aligns with the x -axis. The RIS has a uniform rectangular array (URA) of $N = N_x N_y$ reflecting elements (units), and a controller to dynamically control the phase shift of each unit. Let $\Theta_t := \text{diag}(e^{j\theta_1^t}, \dots, e^{j\theta_N^t})$ be the phase shift matrix for the RIS at time slot t , where $\theta_n^t = \theta_{(n_x-1)N_y+n_y}^t \in [-\pi, \pi]$, $n = 1, \dots, N$, is the phase shift of the n th reflecting unit which is located at the n_y th row and the n_x th column of the RIS, and $j = \sqrt{-1}$. The first element of the RIS is at the right bottom corner of the RIS, and its coordinates are $\mathbf{q}_R = [x_R, y_R, z_R]^T$.

C. Channel Model

The distances of the BS-UAV link d_{BU}^t , the BS-RIS link d_{BR} , the Jammer-RIS link d_{JR} , the Jammer-UAV link d_{JU}^t , and RIS-UAV link d_{RU}^t are given by

$$d_{BU}^t = \|\mathbf{q}_t\| \quad \forall t, \quad d_{BR} = \|\mathbf{q}_R\|, \quad d_{JR} = \|\mathbf{q}_J - \mathbf{q}_R\| \quad (2a)$$

$$d_{JU}^t = \|\mathbf{q}_t - \mathbf{q}_J\|, \quad d_{RU}^t = \|\mathbf{q}_t - \mathbf{q}_R\| \quad \forall t. \quad (2b)$$

Consider an LoS channel for the RIS-UAV link (i.e., the R-U link), and Rician fading channels between the BS/jammer and the UAV (i.e., the B-U and J-U links), and between the BS/jammer and the RIS (i.e., the B-R and J-R links). The channel gains of the B-U and J-U links are: $\forall t$

$$h_{\text{BU}}^t = \sqrt{\rho(d_{\text{BU}}^t)^{-\kappa_1}} \left(\sqrt{\frac{\beta_t}{1+\beta_t}} g_{\text{BU}}^t + \sqrt{\frac{1}{1+\beta_t}} \tilde{g}_{\text{BU}}^t \right) \quad (3a)$$

$$h_{\text{JU}}^t = \sqrt{\rho(d_{\text{JU}}^t)^{-\kappa_1}} \left(\sqrt{\frac{\beta_t}{1+\beta_t}} g_{\text{JU}}^t + \sqrt{\frac{1}{1+\beta_t}} \tilde{g}_{\text{JU}}^t \right) \quad (3b)$$

where ρ stands for the path loss at the reference distance $d_0 = 1$ m with the path loss exponent $\kappa_1 > 2$; β_t is the Rician factor of the B-U and J-U links; g_{BU}^t and g_{JU}^t are the deterministic LoS components with $|g_{\text{BU}}^t| = 1$ and $|g_{\text{JU}}^t| = 1$; $\tilde{g}_{\text{BU}}^t$ and $\tilde{g}_{\text{JU}}^t$ stand for stochastic dispersion captured by a zero-mean, unit-variance circularly symmetric complex Gaussian (CSCG) random variable. The elevation angle reliant Rician factor β_t is captured by the following exponential function [42]:

$$\beta_t = \xi_1 \exp(\xi_2 \arcsin(z_t/d_{\text{BU}}^t)) \quad \forall t \quad (4)$$

where ξ_1 and ξ_2 are two constant coefficients dependent on the environment.

The channel gains of the B-R, J-R, and R-U links, denoted by $\mathbf{h}_{\text{BR}} \in \mathbb{C}^{N \times 1}$, $\mathbf{h}_{\text{JR}} \in \mathbb{C}^{N \times 1}$, and $\mathbf{h}_{\text{RU}}^t \in \mathbb{C}^{N \times 1}$, are

$$\mathbf{h}_{\text{BR}} = \underbrace{\sqrt{\rho d_{\text{BR}}^{-\kappa_2}}}_{\text{path loss}} \underbrace{\left(\sqrt{\frac{\beta}{1+\beta}} \mathbf{h}_{\text{BR}}^{\text{los}} + \sqrt{\frac{1}{1+\beta}} \mathbf{h}_{\text{BR}}^{\text{nlos}} \right)}_{\text{array response \& small-scale fading}} \quad (5a)$$

$$\mathbf{h}_{\text{JR}} = \underbrace{\sqrt{\rho d_{\text{JR}}^{-\kappa_2}}}_{\text{path loss}} \underbrace{\left(\sqrt{\frac{\beta}{1+\beta}} \mathbf{h}_{\text{JR}}^{\text{los}} + \sqrt{\frac{1}{1+\beta}} \mathbf{h}_{\text{JR}}^{\text{nlos}} \right)}_{\text{array response \& small-scale fading}} \quad (5b)$$

$$\mathbf{h}_{\text{RU}}^t = \sqrt{\rho(d_{\text{RU}}^t)^{-2}} \mathbf{g}_{\text{RU}}^t \quad \forall t. \quad (5c)$$

Here, β is the Rician factor of the B-R and J-R links (c.f. β_t); and $\mathbf{h}_{\text{BR}}^{\text{los}}$ and $\mathbf{h}_{\text{JR}}^{\text{los}}$ are the LoS components, as given by

$$\mathbf{h}_{\text{BR}}^{\text{los}} = \begin{bmatrix} 1, \dots, e^{-j\frac{2\pi d_x}{\lambda}(N_x-1)\phi_{\text{BR}}^x} \end{bmatrix}^T \otimes \begin{bmatrix} 1, \dots, e^{-j\frac{2\pi d_y}{\lambda}(N_y-1)\phi_{\text{BR}}^y} \end{bmatrix}^T \quad (6a)$$

$$\mathbf{h}_{\text{JR}}^{\text{los}} = \begin{bmatrix} 1, \dots, e^{-j\frac{2\pi d_x}{\lambda}(N_x-1)\phi_{\text{JR}}^x} \end{bmatrix}^T \otimes \begin{bmatrix} 1, \dots, e^{-j\frac{2\pi d_y}{\lambda}(N_y-1)\phi_{\text{JR}}^y} \end{bmatrix}^T \quad (6b)$$

where d_x and d_y are the antenna spacings in the directions of the x - and y -axes, respectively; $\phi_{\text{BR}}^x = x_R/d_{\text{BR}}$ and $\phi_{\text{BR}}^y = y_R/d_{\text{BR}}$ are the spatial frequencies corresponding to the Angles of Arrivals (AoAs) from BS to RIS, and $\phi_{\text{JR}}^x = (x_J - x_R)/d_{\text{JR}}$ and $\phi_{\text{JR}}^y = (y_J - y_R)/d_{\text{JR}}$ are the spatial frequencies corresponding to the AoAs from the jammer to the RIS along the x - and y -axes, respectively [43].

In (5a) and (5b), $\mathbf{h}_{\text{BR}}^{\text{nlos}} \in \mathbb{C}^{N \times 1}$ and $\mathbf{h}_{\text{JR}}^{\text{nlos}} \in \mathbb{C}^{N \times 1}$ are the Non LoS (NLoS) components with the variables independently drawn from the zero-mean, unit-variance CSCG distribution. In (5c), $\mathbf{g}_{\text{RU}}^t$ is the array response, as given by

$$\mathbf{g}_{\text{RU}}^t = \begin{bmatrix} 1, \dots, e^{-j\frac{2\pi d_x}{\lambda}(N_x-1)\phi_{\text{RU},x}^t} \end{bmatrix}^T \otimes \begin{bmatrix} 1, \dots, e^{-j\frac{2\pi d_y}{\lambda}(N_y-1)\phi_{\text{RU},y}^t} \end{bmatrix}^T \quad \forall t \quad (7)$$

where $\phi_{\text{RU},x}^t = (x_t - x_R)/d_{\text{RU}}^t$ and $\phi_{\text{RU},y}^t = (y_t - y_R)/d_{\text{RU}}^t$ are the spatial frequencies corresponding to the Angles of Departures (AoDs) from RIS to UAV along the x - and y -axes, respectively.

It is noteworthy that the UAV does not ask for the CSI involving the RIS and the jammer to produce its flight path and configure the RIS in this article. Instead, the UAV only measures its own received data rate to learn the control policy of its flight path and RIS configuration. This consideration is of practical value, since the RIS-reflected channels are difficult and slow to estimate.

III. PROBLEM FORMULATION

Let $P_t \forall t$ denote the transmit power of the BS and P_J denote the transmit power of the jammer. The signal-to-interference-plus-noise ratio (SINR) at the UAV at time slot t , denoted by γ_U^t , is

$$\gamma_U^t = \frac{P_t |h_{\text{BU}}^t + (\mathbf{h}_{\text{RU}}^t)^H \boldsymbol{\Theta}_t \mathbf{h}_{\text{BR}}|^2}{P_J |h_{\text{JU}}^t + (\mathbf{h}_{\text{RU}}^t)^H \boldsymbol{\Theta}_t \mathbf{h}_{\text{JR}}|^2 + \sigma_U^2} \quad (8)$$

where σ_U^2 is the variance of the additive white Gaussian noise (AWGN) at the UAV. The received data rate of the UAV at time slot t is given by

$$R_U^t = \log_2(1 + \gamma_U^t). \quad (9)$$

We aim to maximize the total received data rate of the UAV from the BS over the mission duration of T_w slots. The problem considered is stated as follows:

$$\max_{\{\mathbf{q}_t, \boldsymbol{\Theta}_t \forall t\}} \sum_{t=1}^{T_w} R_U^t \quad (10a)$$

$$\text{s.t. } -\pi \leq \theta_n^t < \pi \quad \forall n, t \quad (10b)$$

$$(1a) - (1f).$$

Problem (10) is challenging for traditional convex solvers due to several reasons: first of all, the received data rate is a non-convex function of the UAV's flight path $\mathbf{q}_t \forall t$ and the RIS phase shifts $\boldsymbol{\Theta}_t \forall t$. Second, the UAV flight path waypoints are embedded in the exponents of the R-U link in (5c) and (7), making the trajectory optimization intractable for existing convex tools, such as successive convex approximation. Another reason is that the large number of RIS reflecting units can cause prohibitive overhead and complexity for radio channel estimation, acquisition and reconfiguration. To overcome these limitations, the next section proposes using DRL to solve (10).

IV. PROPOSED DRL FRAMEWORK FOR ANTI-JAMMING COMMUNICATION OF UAV-BORNE IoT PLATFORMS

The proposed method in this section aims to solve problem (10) by utilizing the DDPG and TD3 models. Our approach involves learning to adjust the RIS and control the UAV's trajectory, including heading and acceleration, based on changes in the received data rate of the UAV. Importantly, our method eliminates the need for precise CSI or knowledge of the RIS reflecting channels. DDPG and its variations, such as TD3, have been demonstrated to be effective in addressing problems with continuous action spaces [44], [45], [46]. In

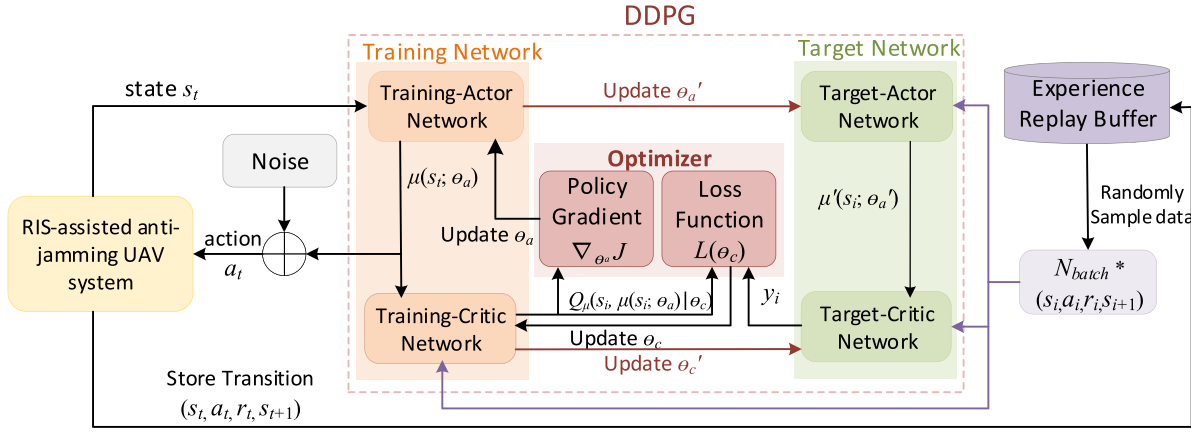


Fig. 2. Proposed DDPG-based framework with a training network and a target network, each comprising an actor network and a critic network. The experience replay buffer gives batches of samples of state transitions for training and updating the networks.

contrast, traditional DRL methods, such as deep Q -learning, can struggle and even diverge when faced with continuous action spaces.

A. State, Action, and Reward

Since the current UAV location only depends on its previous location and speed, the UAV trajectory (i.e., waypoints) is a Markov decision process (MDP). The RIS configuration depends solely on the instantaneous position of the UAV. Therefore, we interpret problem (10) as an MDP with its state, action, and reward defined below.

- 1) *State Space* $\mathcal{S}$: At time slot t , the system state $s_t \in \mathcal{S}$ is made of the relative position of the UAV with regards to its final location, $\mathbf{q}_t - \mathbf{q}_F$, the velocity of the UAV, $\mathbf{V}_t$, and the SINR at the UAV, γ_{U}^t , $s_t = \{\mathbf{q}_t - \mathbf{q}_F, \mathbf{V}_t, \gamma_U^t\}$.
- 2) *Action Space* $\mathcal{A}$: It gathers all possible actions, i.e., $a_t \in \mathcal{A}$. During the t th time step, the action a_t consists of the reflecting coefficients $\{\theta_n^t\}_{n \in \mathcal{N}}$ and the acceleration of the UAV, $\mathbf{A}_t := [A_{xt}, A_{yt}, A_{zt}]^T$, i.e., $a_t = \{\theta_n^t \in [-\pi, \pi] \forall n, A_{it} \in [-A_{\max}, A_{\max}], i \in \{x, y, z\}\}$. The UAV acceleration is constrained by (1d)–(1f). Given the initial location and velocity of the UAV, its future waypoints $\mathbf{q}_t$ and velocities $\mathbf{V}_t$ are decided by the accelerations, i.e., by (1a)–(1c).
- 3) *Reward* r_t : The reward function gives positive returns per time step for implementing action a_t

$$r_t = \underbrace{R_U^t}_{\text{communication}} + \underbrace{\zeta(d_F^{t-1} - d_F^t)}_{\text{distance to the final location}} \quad (11)$$

where R_U^t is the UAV's received data rate, $d_F^{t-1} = \|\mathbf{q}_{t-1} - \mathbf{q}_F\|$ and $d_F^t = \|\mathbf{q}_t - \mathbf{q}_F\|$ are the distances from the UAV to the final location at the $(t-1)$ th and t th time steps, respectively; and ζ is a tunable parameter during the learning process. The second element on the right-hand side of (11) encourages the UAV to fly toward the final location.

- 4) *Policy*: A projection from the state space, $\mathcal{S}$, to the action space $\mathcal{A}$ is referred to as a policy, $\mu : \mathcal{S} \rightarrow \mathcal{A}$,

a distribution $\mu(a|s) = \Pr(a_t = a|s_t = s)$ over state $s \in \mathcal{S}$.

- 5) *Experience*: The experience, defined as $e_t = (s_t, a_t, r_t, s_{t+1})$, is stored in an experience replay memory $\mathbf{R}$.

The UAV experiences state s_t , performs action a_t , receives reward r_t , and turns to state s_{t+1} . A policy $a_t = \mu(s_t)$ maps state s_t to a possible action. The UAV chooses the policy that maximizes the cumulative reward $R_t = \sum_{n=t}^N \gamma^{n-t} r_n$. Here, $\gamma \in (0, 1)$ gives the discount factor. Given s_t , a_t , and μ , the Q -function evaluates R_t by

$$Q_\mu(s_t, a_t) = \mathbb{E}_\mu[R_t | s_t, a_t]. \quad (12)$$

The action-value function, $Q_\mu(s_t, a_t)$, follows the Bellman expectation equation:

$$Q_\mu(s_t, a_t) = \mathbb{E}_{r_t, s_{t+1} \sim \mathcal{E}}[r_t + \gamma \mathbb{E}_{a_{t+1} \sim \mu}[Q_\mu(s_{t+1}, a_{t+1})]]. \quad (13)$$

Here, $\mathcal{E}$ stands for the environment the UAV experiences.

It is generally challenging to directly use an RL algorithm to solve the continuous-space, finite-horizon MDP and determine the Q -value, $Q(s_t, a_t)$, due to the continuous state and action spaces. This article puts forth a new DDPG-based algorithm to control the UAV's trajectory and configure the RIS, as delineated in the following section.

B. Actor-Critic Framework-Based DDPG

The DDPG-based network uses four DNN approximators, including training-actor and training-critic networks, and target-actor and target-critic networks, as shown in Fig. 2. The training-actor network with parameters θ_a , denoted as $\mu(s_t; \theta_a)$, gives an approximate policy of the UAV and produces the actions. The training-critic network with parameters θ_c , denoted as $Q_\mu(s_t, a_t; \theta_c)$, estimates the action-value function concerning the actions created in the training-actor network [44]. The target-actor network with parameter θ'_a , represented by $\mu'(s_t; \theta'_a)$, and the target-critic networks with parameter θ'_c , represented by $Q'_{\mu'}(s_t, a_t; \theta'_c)$, generate the target Q -value for training the training-actor and training-critic networks.

The DDPG network uses the deterministic policy gradient (DPG) theorem [44] to refresh θ_a , θ_c , θ'_a , and θ'_c . It produces actions in an actor-critic setting. Additionally, the adoption of a target network (i.e., the target-actor and target-critic networks) helps prevent unstable learning, as opposed to using only a training network (with a training-actor and a training-critic network) [47].

The UAV inputs state s_t into the training-actor network. Using the DPG theorem [44], the network generates the strategy by projecting the state to an action in a deterministic fashion. This network approximates the agent's policy function and selects action a_t . A noise is added to a_t to balance between new and known actions, resulting in an output action $a_t = \mu(s_t; \theta_a) + \mathcal{N}_t$. Herein, $\mathcal{N}_t$ is a random noise process with a normal distribution. The agent is rewarded with r_t and transitions to state s_{t+1} . Then, it stores the experience (s_t, a_t, r_t, s_{t+1}) in $\mathbf{R}$.

The training-critic network evaluates the action-value function $Q_\mu(s_t, \mu(s_t; \theta_a); \theta_c)$ of the selected action a_t . By using a random sample from the replay memory $\mathbf{R}$, the network approximates the action-value function as $Q_\mu(s_i, \mu(s_i; \theta_a); \theta_c)$. We take $J(\theta_a)$ to be the probability distribution of the parameter θ_a . The training-actor network is adjusted in the direction that improves the strategy the most rapidly, i.e., in the direction of the gradient of $J(\theta_a)$ with respect to (w.r.t.) θ_a [44]

$$\nabla_{\theta_a} J(\theta_a) = \mathbb{E}_{s \sim \rho^\mu} [\nabla_{\theta_a} Q_\mu(s_t, \mu(s_t; \theta_a); \theta_c)] \quad (14a)$$

$$= \mathbb{E}_{s \sim \rho^\mu} [\nabla_{\theta_a} \mu(s_t; \theta_a) \nabla_a Q_\mu(s_t, a; \theta_c)|_{a=\mu(s_t; \theta_a)}] \quad (14b)$$

where (14b) uses the chain rule; ρ^μ provides a discounted state distribution of $\mu(s; \theta_a)$ [45]; $\nabla_{\theta_a} \mu(s)$ gives the gradient of the training-actor network $\mu(s)$ w.r.t. θ_a ; and $\nabla_a Q_\mu(s_t, a; \theta_c)$ provides the gradient of $Q_\mu(s_t, a; \theta_c)$ w.r.t. a .

By randomly drawing N_{batch} sampled historical transitions from $\mathbf{R}$, the gradient $\nabla_{\theta_a} J(\theta_a)$ is approximated by

$$\nabla_{\theta_a} J(\theta_a) \approx \frac{1}{N_{\text{batch}}} \sum_{i=1}^{N_{\text{batch}}} [\nabla_{\theta_a} \mu(s_i) \nabla_a Q_\mu(s_i, a; \theta_c)|_{a=\mu(s_i)}]. \quad (15)$$

The training-actor network parameter, i.e., θ_a , is refreshed based on the gradient ascent [48]

$$\begin{aligned} \theta_a &\leftarrow \theta_a + \eta_a \nabla_{\theta_a} J(\theta_a) \\ &\approx \theta_a + \frac{\eta_a}{N_{\text{batch}}} \sum_{i=1}^{N_{\text{batch}}} [\nabla_{\theta_a} \mu(s_i) \nabla_a Q_\mu(s_i, a; \theta_c)|_{a=\mu(s_i)}]. \end{aligned} \quad (16)$$

Here, η_a specifies the learning rate of the training-actor network.

The training-critic network is refreshed through minimizing the following loss function:

$$L(\theta_c) = \mathbb{E}_{s_t \sim \rho^\mu, a_t \sim \mu(s_t; \theta_a)} [(Q_\mu(s_t, a_t; \theta_c) - y_t)^2]. \quad (17)$$

Here, $y_t = r_t + \gamma Q'_{\mu'}(s_{t+1}, \mu'(s_{t+1}; \theta'_a); \theta'_c)$ is the target Q -value produced by the target network under the transition

(s_t, a_t, r_t, s_{t+1}) . Here, the parameters of the target-actor and target-critic networks, θ'_a and θ'_c , are the respective decayed copies of θ_a and θ_c .

With N_{batch} randomly sampled transitions, the loss function, $L(\theta_c)$, is approximately evaluated by

$$L(\theta_c) \approx \frac{1}{N_{\text{batch}}} \sum_{i=1}^{N_{\text{batch}}} [(Q_\mu(s_i, \mu(s_i; \theta_a); \theta_c) - y_i)^2] \quad (18)$$

where $y_i = r_i + \gamma Q'_{\mu'}(s_{i+1}, \mu'(s_{i+1}; \theta'_a); \theta'_c)$ gives the approximate target Q -value that the target network generates upon N_{batch} transitions sampled at random. By differentiating $L(\theta_c)$ w.r.t. θ_c , the gradient is attained

$$\begin{aligned} \nabla_{\theta_c} L(\theta_c) &\approx \frac{2}{N_{\text{batch}}} \sum_{i=1}^{N_{\text{batch}}} [(Q_\mu(s_i, \mu(s_i; \theta_a); \theta_c) - y_i) \\ &\quad \times \nabla_{\theta_c} Q_\mu(s_i, \mu(s_i; \theta_a); \theta_c)]. \end{aligned} \quad (19)$$

The training-critic network parameter, θ_c , is refreshed by utilizing the stochastic gradient descent method [48].

The target-actor and target-critic networks are refreshed based on the training-actor and training-critic networks

$$\begin{aligned} \theta'_a &\leftarrow \tau_a \theta_a + (1 - \tau_a) \theta'_a \\ \theta'_c &\leftarrow \tau_c \theta_c + (1 - \tau_c) \theta'_c \end{aligned} \quad (20)$$

where τ_a and τ_c are the decaying rates for the training-actor and training-critic networks, respectively.

C. Twin Delayed DDPG

TD3 is one of the latest extensions of DDPG and consists of a training network and a target network, where the training network is made of a training-actor and two training-critic networks, and the target network comprises a target-actor and two target-critic networks, as shown in Fig. 3. TD3 addresses the Q -value overestimation problem of the DDPG algorithm by incorporating *three improvements* over the classical DDPG model, namely, *clipped double- Q learning*, *target policy smoothing*, and *delayed policy update* [49], [50].

- 1) *Clipped Double- Q Learning*: TD3 contains two training-critic and target-critic networks to produce two Q -values. The lesser of the two is used to evaluate the target Q -value in the Bellman error loss function. Specifically, $Q_\mu(s_t, a_t; \theta_c)$ in the DDPG is replaced by $Q_\mu(s_t, a_t) = \min\{Q_1(s_t, a_t; \theta_1), Q_2(s_t, a_t; \theta_2)\}$ in the TD3.
- 2) *Target Policy Smoothing*: TD3 perturbs actions produced by the target-actor network (i.e., “target action”) with noises and smooths the corresponding Q -function values to enhance the resistance of the policy against erroneous Q -functions. The smoothed target action is written as

$$a'_t = \text{clip}(\mu'(s_{t+1}; \theta'_a) + \text{clip}(\epsilon', -\sigma_m^2, \sigma_m^2), a_{\min}, a_{\max}). \quad (21)$$

Here, the noise ϵ' is taken at random from a Gaussian distribution with zero mean and variance σ_a^2 , i.e., $\epsilon' \sim \mathcal{N}(0, \sigma_a^2)$; and σ_m^2 is the maximum exploration noise supported by the environment.

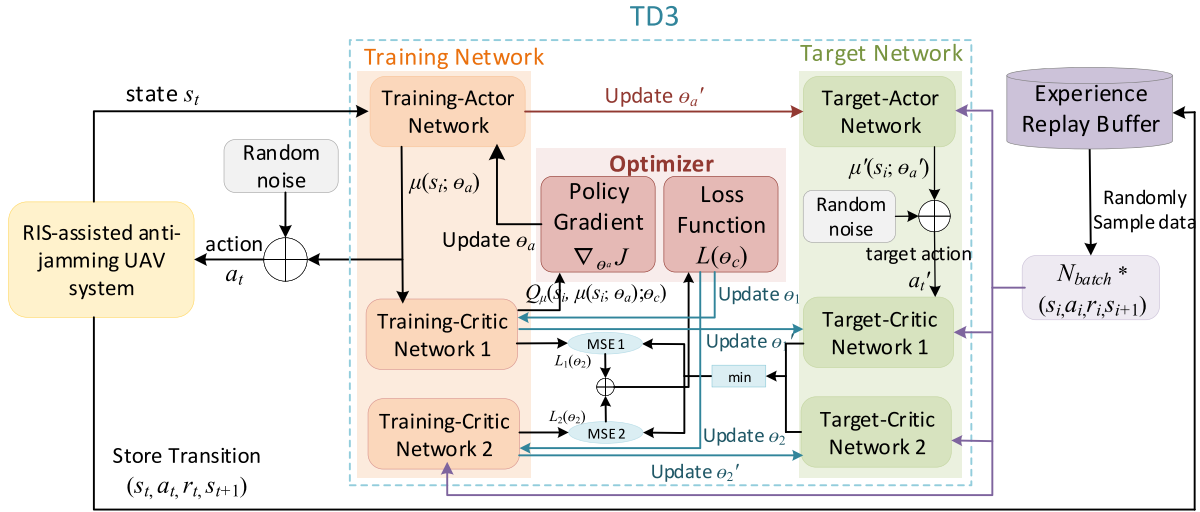


Fig. 3. Proposed TD3-based framework with an actor network comprising an actor and a target-actor, and a critic network comprising two critics and two target critics. The experience replay buffer gives batches of samples of state transitions for training and updating the networks.

TABLE III
PARAMETERS OF THE SYSTEM MODEL

Parameter	Value
RIS antenna separation, d_x, d_y	$\lambda/2$
Maximum and minimum speeds, $V_{\max}, V_{\min}$	40 m/s, 2 m/s
Path loss, ρ	-30 dB
Path loss exponents, κ_1, κ_2	3.5, 2.8
Rician factor coefficients, ξ_1, ξ_2	1, 4.4
Rician factor, β	3 dB
Noise power, σ_U^2	-169 dBm

3) “Delayed” Policy Update: The training-actor and target-actor networks (i.e., policies) are refreshed less frequently than the training-critic and target-critic networks. For example, it was recommended in [49] that the training-actor and target-actor networks are refreshed after the training-critic and target-critic networks are refreshed twice in TD3.

In contrast, the DDPG model only contains a training-critic network and a target-critic network, and produces a Q -value. The DDPG model does not add noises toward target actions. Moreover, the classical DDPG model refreshes its train-actor and target-actor networks and train-critic and target-critic networks at the same pace.

V. PERFORMANCE EVALUATION

We carry out extensive experiments in Python to evaluate the proposed approach. The location of the jammer is $\mathbf{q}_J = [-25, -25, 0]^T$ m. The UAV’s initial and final locations are $\mathbf{q}_0 = [200, -100, 5]^T$ m and $\mathbf{q}_F = [100, 60, 50]^T$ m. The RIS has $N = 5 \times 4 = 20$ (or $N = 5 \times 8 = 40$) reflecting elements, and the reference point is $\mathbf{q}_R = [50, 50, 30]^T$ m. The scheduling horizon is $T = 30$ s with each time slot being $\delta = 0.1$ s. The other parameters concerning the system model are collated in Table III.

A. Experiment Settings

The proposed DDPG network is composed of actor networks implemented using fully connected neural networks

(FCNNs) with three hidden layers and learning rates of 10^{-4} . The first, second, and third layers of the actor networks have 64, 128, and 64 neurons, respectively. The output layer implements the $\tanh(\cdot)$ activation function to bound the output actions within $[-\pi, \pi)$ for the RIS configuration and $[-2, 2]$ m/s² for the UAV control. Additionally, this article utilizes critic networks that employ FCNNs with two hidden layers and learning rates of 10^{-3} . Both hidden layers utilize the rectified linear unit (ReLU) activation functions with 64 neurons in the first layer and 128 neurons in the second layer. The DDPG actor policy is trained using additive noise $\mathcal{N}$, which is sampled from a complex Gaussian noise distribution with zero mean and variance 0.2.

The proposed TD3 network is built upon the DDPG network. It includes two duplicates of the training-critic and target-critic networks; see Fig. 3. Similar to the DDPG network, the actor in the TD3 network is trained using exploration noise that is drawn from a complex Gaussian distribution with zero mean and variance 0.2. Additionally, the target-actor in the TD3 network is smoothed using policy noise that is drawn from a complex Gaussian distribution with zero mean and variance of 0.2. The maximum exploration noise is set to 0.5, and the actor networks are refreshed every two steps. The TD3 improves the DDPG by providing faster and smoother convergence, which is especially beneficial for larger RISs.

The hyperparameters of the proposed DDPG and TD3 networks are summarized in Table IV. The DDPG and TD3 networks are trained on a server equipped with a NVIDIA Tesla P100 SXM2 16GB GPU.

As discussed in Section I, no existing works have addressed the RIS-assisted jamming rejection for a UAV-borne IoT platform. For this reason, it may not be fair or adequate to compare the proposed approach in this work with those existing works. With due diligence, we come up with two baselines to further verify the merits of the proposed approach.

Baseline 1: This baseline applies TD3 to UAV flight path planning in the absence of the RIS, referred to as “baseline

TABLE IV
HYPERPARAMETERS OF THE PROPOSED DDPG AND TD3 ALGORITHMS

Parameter	Value
Reduction coefficient for upcoming reward, γ	0.99
Training coefficient for actor and critic networks, η_a, η_c	1×10^{-4}
Declining coefficient for actor and critic networks, $\tau_a, \tau_c, \rho_\tau$	5×10^{-3}
Capacity for experience repetition	1×10^5
Quantity of episodes, T_{ep}	3000
Total steps per episodes, T_s	300
Quantity of experiences in a mini-batch, N_{batch}	128
Variance of the exploration noise, σ_e^2	0.2
Delayed policy update interval (TD3)	2
Variance of the policy noise (TD3), σ_a^2	0.2
Largest value of the Gaussian noise (TD3), σ_m^2	0.5

without RIS” or simply “without RIS.” The TD3 algorithm learns the UAV’s trajectory solely based on the UAV’s received data rate without CSI involving the RIS or jammer.

Baseline 2: This baseline decouples the UAV’s trajectory plan from the RIS configuration by first using a TD3-based algorithm to optimize the UAV’s trajectory given the RIS configuration, and then maximizing the signal-to-noise ratio (SNR) at each time slot using the Dinkelbach method under the assumption of perfect and instantaneous CSI for all involved channels. The Dinkelbach method is used to reformulate the SNR maximization problem as a fractional program defined as $F(\gamma_U^t) = \min_{\Theta_t} f(\Theta_t) - \gamma_U^t g(\Theta_t)$, s.t. $-\pi \leq \theta_n^t < \pi \forall n, t$, where $\gamma_U^t = [f(\Theta_t)/g(\Theta_t)]$ and Θ_t gives the RIS configuration. Given γ_U^t , the fractional program can be reorganized as a quadratic program with a unit-modulus constraint and solved using manifold optimization. The value of γ_U^t is refreshed based on the resultant Θ_t . This process is repeated until convergence, and the convergent value of γ_U^t is output [51].

B. Results of Policy Learning

Fig. 4 plots the per episode and average rewards of the proposed and baseline algorithms for $N = 20$ and $N = 40$. The average reward for the i th training episode, denoted by $\bar{r}_i$, is evaluated as $\bar{r}_i = (1/i) \sum_{j=1}^i r_j$, where $i = 1, \dots, T_{ep}$, and r_j is the step reward for the j th training episode; see (11). The per-episode reward indicates the reward obtained by the agent in a single episode of the task, while the average reward accounts for the cumulative reward over multiple episodes. The average reward is useful when the reward function is noisy or stochastic, because the per-episode reward may fluctuate significantly over episodes, making it hard to accurately evaluate the agent’s performance. By taking the average over multiple episodes, we can get a more stable estimate of the agent’s performance.

Fig. 4 shows that the average reward gradually increases, as the UAV control policy adapts to a randomly generated target trajectory in each episode. The proposed TD3 algorithm outperforms the baselines, and achieves its maximum reward at the 654th episode for $N = 20$, and at the 477th episode for $N = 40$. The proposed DDPG algorithm reaches its maximum reward at the 1607th episode when $N = 20$, and at the 2995th when $N = 40$. Baseline 1 without RIS reaches its maximum reward at the 280th episode. Baseline 2 reaches its maximum reward at the 1752nd episode when $N = 20$. The fast convergence of Baseline 2 is due to its substantially

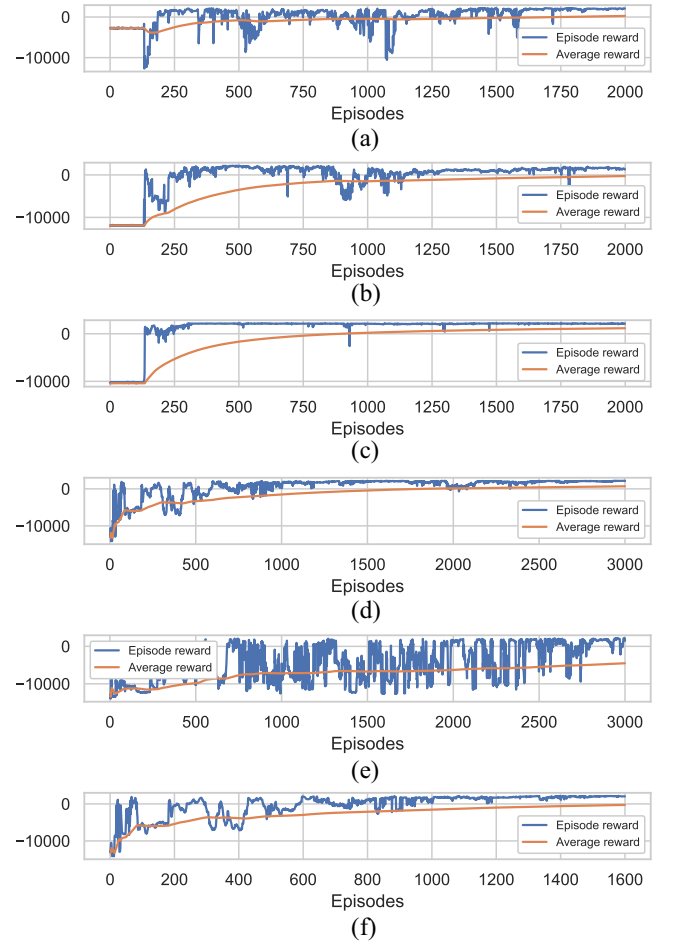


Fig. 4. Per episode and average rewards of the proposed DDPG and TD3 algorithms: (a) plots the proposed TD3 scheme when $N = 20$; (b) plots the proposed TD3 scheme when $N = 40$; (c) plots Baseline 2 when $N = 20$; (d) plots the proposed DDPG scheme when $N = 20$; (e) plots the proposed DDPG scheme when $N = 40$; and (f) plots Baseline 1 “without RIS.”

smaller action space of only UAV accelerations resulting from an unrealistic assumption of perfect and instantaneous CSI of all involved channels. In general, the TD3 converges faster and more smoothly than the DDPG. Yet, it undergoes less smooth changes in the per-episode reward when the action space is smaller, i.e., $N = 20$. This is because the TD3 has two critic networks for both the training and target networks (c.f. Fig. 3), which could incur higher complexity and more randomness when training, especially when the action space is smaller. In contrast, DDPG is suitable for training tasks that are less complicated and have relatively smaller action spaces.

C. Test Results of Learned Policy

Using the learning results obtained in Section V-B, we test the proposed DDPG-based and TD3-based algorithms for $N = 20$ and 40 at the RIS, as well as the baseline schemes for $N = 20$ at the RIS for comparison. 500 testing episodes are conducted, each consisting of 300 steps. During testing, no exploration noise is added. The proposed algorithms and baselines are evaluated in terms of the 3-D UAV trajectory and received data rate. Fig. 5 shows the 3-D trajectory of the UAV,

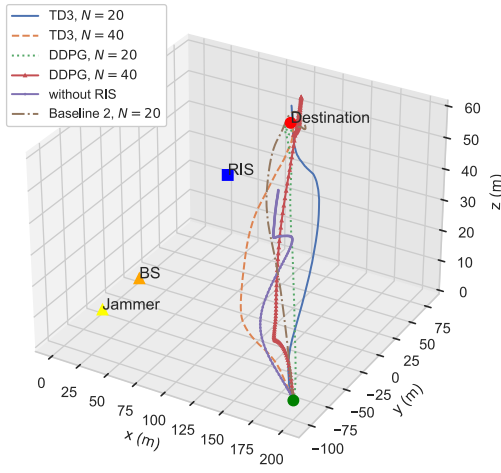


Fig. 5. 3-D UAV trajectory, where the green and red dots are the UAV initial location and its expected destination, the orange and yellow triangles denote the locations of the BS and Jammer, and the blue square is the RIS reference point.

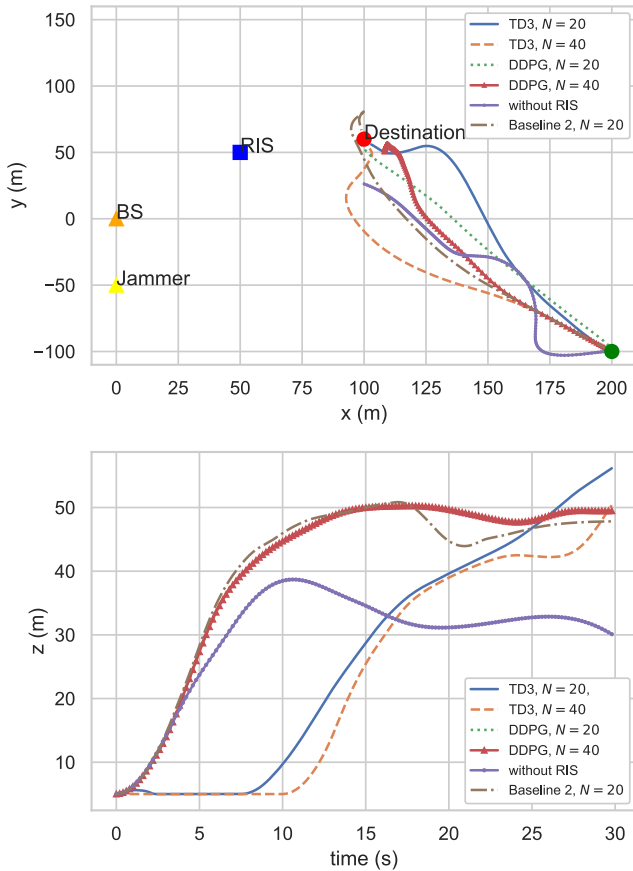


Fig. 6. Projection of the UAV trajectory on the x - y plane, with the green and red dots being the UAV initial and final locations, the orange and yellow triangles being the locations of the BS and Jammer, and the blue square being the RIS reference point.

while Fig. 6 illustrates the trajectory in the x - y plane and along the z -axis. These figures demonstrate that the UAV is able to adapt its control policy to the anti-jamming communication and successfully reach the destination.

Fig. 7 shows the received data rate of the UAV as the mission duration increases. The results are based on an average of

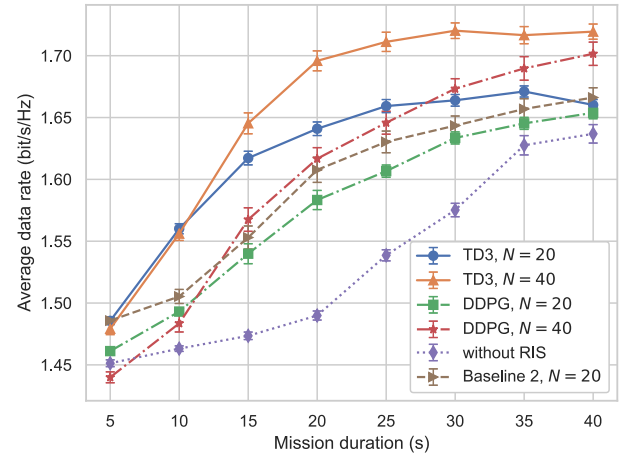


Fig. 7. Received data rate of the UAV versus the mission duration averaged over 500 independent testing runs.

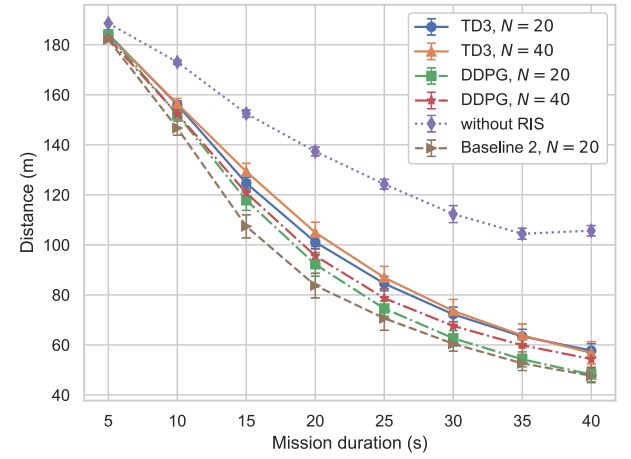


Fig. 8. Finishing distance between the UAV and its destination at the end of the mission with the increase of the mission duration.

500 independent testing episodes, with error bars representing the associated uncertainty. It is observed that the received data rate increases with the mission duration under all considered algorithms, and the use of an RIS improves the received data rate. The proposed TD3 and DDPG algorithms give slightly lower data rates than Baseline 2, but operate without the CSI involving the RIS or the jammer, demonstrating their ability to adapt to system changes. Additionally, the proposed TD3 algorithm achieves a substantially higher data rate than the proposed DDPG algorithm, particularly for larger numbers of RIS elements. This is attributed to the faster convergence and better convergent control policy of TD3 compared to DDPG, as previously shown in Fig. 4.

Fig. 8 illustrates the distance between the UAV and its expected destination at the end of the mission. The results show that as the mission duration increases, the distance decreases for all algorithms. Without the use of the RIS, the UAV is unable to reach its expected final location as it must remain close to the BS to maintain a sufficient data rate. However, the proposed algorithms, such as TD3, enable the UAV to get closer or reach its final location by adjusting the RIS to enhance the desired signals, weaken the jamming signals and extend the effective BS-UAV transmission range.

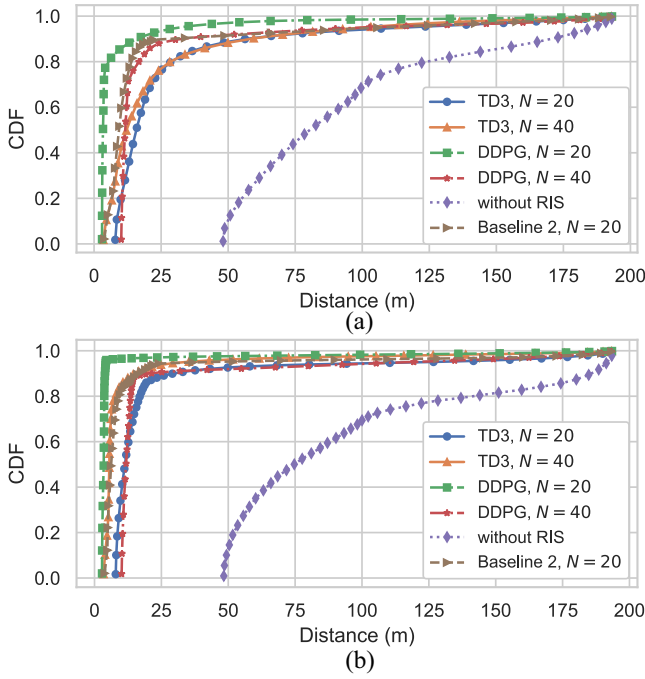


Fig. 9. CDF of the distance from the UAV to the final location when the mission duration is $T = 30$ and 40 s.

It is worth noting that as the mission duration increases, DDPG can increasingly approach TD3 in the average achievable data rate. This indicates that DDPG is suitable for a less constrained problem setting where there is sufficient time for the UAV to maneuver and explore its action space. In this case, DDPG can be a suitable solution, as it can benefit from its simpler network architecture than TD3. In contrast, TD3 demonstrates significant gains over DDPG when the time constraint is more stringent. In other words, TD3 suits better under a shorter mission duration, since it has a more complex network structure and can generate more randomness to test the action space more extensively for better solutions.

Fig. 9 plots the cumulative distribution function (CDF) of the distance between the UAV and its expected destination at the end of the mission, for mission durations of $T = 30$ and 40 s. The results show that as the number of RIS elements (i.e., N) or the mission duration (i.e., T) increases, the proposed TD3 algorithm can get closer to or reach the destination more frequently, and performs significantly better than the case without an RIS. While perfect CSI is important for UAV trajectory planning, as seen in Baseline 2 for $N = 20$, the TD3 algorithm can produce equally effective trajectories without CSI by utilizing a larger RIS with more elements (as seen in TD3 for $N = 40$). On the other hand, the DDPG algorithm appears to suffer from overfitting, as the distance between the UAV and its expected destination has little dispersion. This indicates that DDPG is more prone to overestimating the Q -value function for a small number of possible actions, leading to a noisy gradient for policy refreshes and less effective UAV trajectories and lower data rates.

Fig. 10 plots the received data rate of the UAV versus the finishing distance of the UAV to the destination at the end

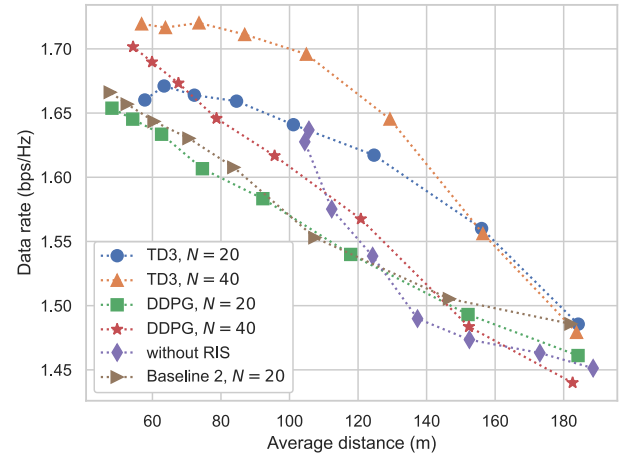


Fig. 10. UAV's data rate versus its distance to the expected destination, where T ranges from 5 to 40 s.

of the mission time. The figure is produced by collating the data rates shown in Fig. 7 and their corresponding finishing distances shown in Fig. 8. Clearly, the values at the top left corner of Fig. 10 indicate a better tradeoff between the data rate and the finishing distance, where the mission duration is 25 to 40 s, as discussed in Figs. 7 and 8. Note that the use of target networks and the noise added when exploring new actions for the DDPG and TD3 algorithms may incur more randomness and lead to an unstable performance during training, as observed in Fig. 10.

VI. CONCLUSION

This article developed a new DRL-driven framework for the trajectory planning and RIS-assisted jamming rejection for a UAV-borne IoT platform. The DDPG model and its enhancement, TD3, were designed to allow the UAV to learn its trajectory and the RIS configuration only based on its received data rate, eliminating the need of CSI for learning. Extensive simulations showed that the proposed DRL algorithms offer the UAV reliable resistance against jamming. The TD3 algorithm converges faster and more smoothly than the DDPG algorithm. It also demonstrates robustness against different locations of the jammer. This is particularly important due to the difficulty in locating the jammer in practice.

REFERENCES

- [1] B. Bera, A. K. Das, S. Garg, M. J. Piran, and M. S. Hossain, "Access control protocol for battlefield surveillance in drone-enabled IoT environment," *IEEE Internet Things J.*, vol. 9, no. 4, pp. 2708–2721, Feb. 2022.
- [2] L. Liu, A. Wang, G. Sun, and J. Li, "Multiobjective optimization for improving throughput and energy efficiency in UAV-enabled IoT," *IEEE Internet Things J.*, vol. 9, no. 20, pp. 20763–20777, Oct. 2022.
- [3] K. Li, W. Ni, X. Wang, R. P. Liu, S. S. Kanhere, and S. Jha, "Energy-efficient cooperative relaying for unmanned aerial vehicles," *IEEE Trans. Mobile Comput.*, vol. 15, no. 6, pp. 1377–1386, Jun. 2016.
- [4] Z. Yu, Y. Gong, S. Gong, and Y. Guo, "Joint task offloading and resource allocation in UAV-enabled mobile edge computing," *IEEE Internet Things J.*, vol. 7, no. 4, pp. 3147–3159, Apr. 2020.
- [5] S. Hu, W. Ni, X. Wang, A. Jamalipour, and D. Ta, "Joint optimization of trajectory, propulsion, and thrust powers for covert UAV-on-UAV video tracking and surveillance," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 1959–1972, 2021.

- [6] X. Yuan, Z. Feng, W. Ni, Z. Wei, R. P. Liu, and J. A. Zhang, "Secrecy rate analysis against aerial eavesdropper," *IEEE Trans. Commun.*, vol. 67, no. 10, pp. 7027–7042, Oct. 2019.
- [7] S. Hu, Q. Wu, and X. Wang, "Energy management and trajectory optimization for UAV-enabled legitimate monitoring systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 142–155, Jan. 2021.
- [8] Y. Dang, C. Benzaïd, B. Yang, T. Taleb, and Y. Shen, "Deep-ensemble-learning-based GPS spoofing detection for cellular-connected UAVs," *IEEE Internet Things J.*, vol. 9, no. 24, pp. 25068–25085, Dec. 2022.
- [9] K. Li, R. C. Voicu, S. S. Kanhere, W. Ni, and E. Tovar, "Energy efficient legitimate wireless surveillance of UAV communications," *IEEE Trans. Veh. Technol.*, vol. 68, no. 3, pp. 2283–2293, Mar. 2019.
- [10] H. Lei et al., "Safeguarding UAV IoT communication systems against randomly located eavesdroppers," *IEEE Internet Things J.*, vol. 7, no. 2, pp. 1230–1244, Feb. 2020.
- [11] Z. Na, Y. Liu, J. Shi, C. Liu, and Z. Gao, "UAV-supported clustered NOMA for 6G-enabled Internet of Things: Trajectory planning and resource allocation," *IEEE Internet Things J.*, vol. 8, no. 20, pp. 15041–15048, Oct. 2021.
- [12] X. Yuan et al., "Secrecy performance of terrestrial radio links under collaborative aerial eavesdropping," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 604–619, 2020.
- [13] B. Duo, Q. Wu, X. Yuan, and R. Zhang, "Anti-jamming 3D trajectory design for UAV-enabled wireless sensor networks under probabilistic LoS channel," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 16288–16293, Dec. 2020.
- [14] J. Peng, Z. Zhang, Q. Wu, and B. Zhang, "Anti-jamming communications in UAV swarms: A reinforcement learning approach," *IEEE Access*, vol. 7, pp. 180532–180543, 2019.
- [15] Z. Li, Y. Lu, X. Li, Z. Wang, W. Qiao, and Y. Liu, "UAV networks against multiple maneuvering smart jamming with knowledge-based reinforcement learning," *IEEE Internet Things J.*, vol. 8, no. 15, pp. 12289–12310, Aug. 2021.
- [16] C. Huang et al., "Holographic MIMO surfaces for 6G wireless networks: Opportunities, challenges, and trends," *IEEE Wireless Commun.*, vol. 27, no. 5, pp. 118–125, Oct. 2020.
- [17] Y. Cao, T. Lv, Z. Lin, and W. Ni, "Delay-constrained joint power control, user detection and passive beamforming in intelligent reflecting surface-assisted uplink mmWave system," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 2, pp. 482–495, Jun. 2021.
- [18] C. Sun, W. Ni, Z. Bu, and X. Wang, "Energy minimization for intelligent reflecting surface-assisted mobile edge computing," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6329–6344, Aug. 2022.
- [19] W. Tang et al., "Wireless communications with reconfigurable intelligent surface: Path loss modeling and experimental measurement," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 421–439, Jan. 2021.
- [20] Q. Wu, S. Zhang, B. Zheng, C. You, and R. Zhang, "Intelligent reflecting surface-aided wireless communications: A tutorial," *IEEE Trans. Commun.*, vol. 69, no. 5, pp. 3313–3351, May 2021.
- [21] Y. Liu et al., "Reconfigurable intelligent surfaces: Principles and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1546–1577, 3rd Quart., 2021.
- [22] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.
- [23] Q. Wu and R. Zhang, "Beamforming optimization for wireless network aided by intelligent reflecting surface with discrete phase shifts," *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1838–1851, Mar. 2020.
- [24] S. Xu, J. Liu, and Y. Cao, "Intelligent reflecting surface empowered physical-layer security: Signal cancellation or jamming?" *IEEE Internet Things J.*, vol. 9, no. 2, pp. 1265–1275, Jan. 2022.
- [25] H. Yang et al., "Intelligent reflecting surface assisted anti-jamming communications: A fast reinforcement learning approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1963–1974, Mar. 2021.
- [26] Z. Peng, Z. Zhang, L. Kong, C. Pan, L. Li, and J. Wang, "Deep reinforcement learning for RIS-aided multiuser full-duplex secure communications with hardware impairments," *IEEE Internet Things J.*, vol. 9, no. 21, pp. 21121–21135, Nov. 2022.
- [27] S. Li, B. Duo, X. Yuan, Y.-C. Liang, and M. Di Renzo, "Reconfigurable intelligent surface assisted UAV communication: Joint trajectory design and passive beamforming," *IEEE Wireless Commun. Lett.*, vol. 9, no. 5, pp. 716–720, May 2020.
- [28] S. Li, B. Duo, M. D. Renzo, M. Tao, and X. Yuan, "Robust secure UAV communications with the aid of reconfigurable intelligent surfaces," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6402–6417, Oct. 2021.
- [29] A. Ranjha and G. Kaddoum, "URLLC facilitated by mobile UAV relay and RIS: A joint design of passive beamforming, blocklength, and UAV positioning," *IEEE Internet Things J.*, vol. 8, no. 6, pp. 4618–4627, Mar. 2021.
- [30] X. Liu, Y. Liu, and Y. Chen, "Machine learning empowered trajectory and passive beamforming design in UAV-RIS wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 7, pp. 2042–2055, Jul. 2021.
- [31] N. Agrawal, A. Bansal, K. Singh, and C.-P. Li, "Performance evaluation of RIS-assisted UAV-enabled vehicular communication system with multiple non-identical interferers," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 9883–9894, Jul. 2022.
- [32] Y. Xu, T. Zhang, Y. Liu, D. Yang, L. Xiao, and M. Tao, "Computation capacity enhancement by joint UAV and RIS design in IoT," *IEEE Internet Things J.*, vol. 9, no. 20, pp. 20590–20603, Oct. 2022.
- [33] X. Pan, Z. Zhang, H. Lin, and J. Zhang, "RIS relaying UAV-aided WPCN for throughput maximization," in *Proc. 6th World Conf. Comput. Commun. Technol. (WCCCT)*, Jan. 2023, pp. 100–105.
- [34] S. K. Singh, K. Agrawal, K. Singh, C.-P. Li, and Z. Ding, "NOMA enhanced hybrid RIS-UAV-assisted full-duplex communication system with imperfect SIC and CSI," *IEEE Trans. Commun.*, vol. 70, no. 11, pp. 7609–7627, Nov. 2022.
- [35] D. Tyrovolas, P.-V. Mekikis, S. A. Tegos, P. D. Diamantoulakis, C. K. Liaskos, and G. K. Karagiannidis, "Energy-aware design of UAV-mounted RIS networks for IoT data collection," *IEEE Trans. Commun.*, vol. 71, no. 2, pp. 1168–1178, Feb. 2023.
- [36] K. Zhao, H. Mei, S. Lyu, and L. Peng, "Joint optimization of multiple UAV-mounted RISs deployment and RIS elements allocation," in *Proc. 13th Int. Conf. Inf. Commun. Technol. Conver. (ICTC)*, Oct. 2022, pp. 1193–1197.
- [37] B.-R. Wu and L.-C. Wang, "Learning from UAV experimental results for performance modeling of reconfigurable intelligent surface flying platform," in *Proc. 30th Wireless Opt. Commun. Conf. (WOCC)*, 2021, pp. 245–250.
- [38] T. Roy, D. Meena, and L. G. M. Prakasam, "FPGA based digital beam forming for radars," in *Proc. IEEE Radar Conf.*, 2009, pp. 1–5.
- [39] J. Cong, B. Liu, S. Neuendorffer, J. Noguera, K. Vissers, and Z. Zhang, "High-level synthesis for FPGAs: From prototyping to deployment," *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.*, vol. 30, no. 4, pp. 473–491, Apr. 2011.
- [40] X. Mu, Y. Liu, L. Guo, J. Lin, and H. V. Poor, "Intelligent reflecting surface enhanced multi-UAV NOMA networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 10, pp. 3051–3066, Oct. 2021.
- [41] C. Sun, W. Ni, and X. Wang, "Joint computation offloading and trajectory planning for UAV-assisted edge computing," *IEEE Trans. Wireless Commun.*, vol. 20, no. 8, pp. 5343–5358, Aug. 2021.
- [42] C. You and R. Zhang, "3D trajectory optimization in Rician fading for UAV-enabled data harvesting," *IEEE Trans. Wireless Commun.*, vol. 18, no. 6, pp. 3192–3207, Jun. 2019.
- [43] H. Lu, Y. Zeng, S. Jin, and R. Zhang, "Aerial intelligent reflecting surface: Joint placement and passive beamforming design with 3D beam flattening," *IEEE Trans. Wireless Commun.*, vol. 20, no. 7, pp. 4128–4143, Jul. 2021.
- [44] D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra, and M. Riedmiller, "Deterministic policy gradient algorithms," in *Proc. 31st Int. Conf. Mach. Learn.*, vol. 32, 2014, pp. 387–395.
- [45] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," in *Proc. ICLR (Poster)*, 2016, pp. 1–10. [Online]. Available: <http://arxiv.org/abs/1509.02971>
- [46] S. Hu, X. Chen, W. Ni, E. Hossain, and X. Wang, "Distributed machine learning for wireless communication networks: Techniques, architectures, and applications," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1458–1493, 3rd Quart., 2021.
- [47] Y. Hou, L. Liu, Q. Wei, X. Xu, and C. Chen, "A novel DDPG method with prioritized experience replay," in *Proc. IEEE Int. Conf. Syst. Man Cybern. (SMC)*, 2017, pp. 316–321.
- [48] R. S. Sutton, D. A. McAllester, S. P. Singh, and Y. Mansour, "Policy gradient methods for reinforcement learning with function approximation," in *Proc. NIPS*, vol. 99, 1999, pp. 1057–1063.

- [49] S. Dankwa and W. Zheng, "Twin-delayed DDPG: A deep reinforcement learning technique to model a continuous movement of an intelligent robot agent," in *Proc. 3rd Int. Conf. Vis. Image Signal Process.*, 2019, pp. 1–5.
- [50] X. Yuan, S. Hu, W. Ni, R.-P. Liu, and X. Wang, "Joint user, channel, modulation-coding selection, and RIS configuration for jamming resistance in multiuser OFDMA systems," *IEEE Trans. Commun.*, vol. 71, no. 3, pp. 1631–1645, Mar. 2023.
- [51] W. Dinkelbach, "On nonlinear fractional programming," *Manag. Sci.*, vol. 13, no. 7, pp. 492–498, 1967.



Shuyan Hu (Member, IEEE) received the B.Eng. degree in electrical engineering from Tongji University, Shanghai, China, in 2014, and the Ph.D. degree in electronic science and technology from Fudan University, Shanghai, in 2019.

She is currently a Postdoctoral Research Fellow with the School of Information Science and Technology, Fudan University. She was selected by Shanghai Postdoctoral Excellence Program in 2019. Her research interests include machine learning and convex optimizations and their applications

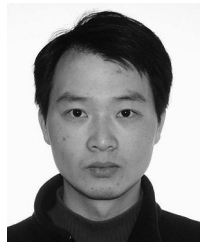
to unmanned aerial vehicle (UAV) networks and intelligent systems.



Xin Yuan (Member, IEEE) received the B.E. degree from Taiyuan University of Technology, Taiyuan, Shanxi, China, in 2013, and the dual Ph.D. degrees from Beijing University of Posts and Telecommunications, Beijing, China, and the University of Technology Sydney, Sydney, NSW, Australia, in 2019 and 2020, respectively.

She is currently a Research Scientist with Commonwealth Scientific and Industrial Research Organization, Sydney. Her research interests include machine learning and optimization, and their applications

to UAV networks and intelligent systems.



Wei Ni (Senior Member, IEEE) received the B.E. and Ph.D. degrees in electronic engineering from Fudan University, Shanghai, China, in 2000 and 2005, respectively.

He is currently a Principal Research Scientist with Commonwealth Scientific and Industrial Research Organization, Sydney, NSW, Australia. He is also a Conjoint Professor with the University of New South Wales, Sydney, an Adjunct Professor with the University of Technology Sydney, Sydney, and a Honorary Professor with Macquarie University,

Sydney. He was a Postdoctoral Research Fellow with Shanghai Jiaotong University, Shanghai, from 2005 to 2008; a Deputy Project Manager with the Bell Labs, Alcatel/Alcatel-Lucent, Shanghai, from 2005 to 2008; and a Senior Researcher with Devices Research and Development, Nokia, Shanghai, from 2008 to 2009. He has authored five book chapters, more than 250 journal papers, more than 100 conference papers, 25 patents, and ten standard proposals accepted by IEEE. His research interests include machine learning, online learning, stochastic optimization, and their applications to system efficiency and integrity.

Dr. Ni is the Chair of the IEEE VTS NSW Chapter since 2020 and an Editor of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS since 2018. He served first as the Secretary and then the Vice Chair for the IEEE VTS NSW Chapter from 2015 to 2019, a Track Chair for VTC-Spring 2017, a Track Co-Chair for IEEE VTC-Spring 2016, a Publication Chair for BodyNet 2015, and a Student Travel Grant Chair for WPMC 2014.



Xin Wang (Fellow, IEEE) received the B.Sc. degree in electrical engineering and the M.Sc. degree in electrical engineering from Fudan University, Shanghai, China, in 1997 and 2000, respectively, and the Ph.D. degree in electrical engineering from Auburn University, Auburn, AL, USA, in 2004.

From September 2004 to August 2006, he was a Postdoctoral Research Associate with the Department of Electrical and Computer Engineering, University of Minnesota, Minneapolis, MN, USA. In August 2006, he joined the Department of

Electrical Engineering, Florida Atlantic University, Boca Raton, FL, USA, as an Assistant Professor, then was promoted to a Tenured Associate Professor in 2010. He is currently a Distinguished Professor and the Chair of the Department of Communication Science and Engineering, Fudan University. His research interests include stochastic network optimization, energy-efficient communications, cross-layer design, and signal processing for communications.

Dr. Wang is a Senior Area Editor for the IEEE TRANSACTIONS ON SIGNAL PROCESSING and an Editor for the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, and in the past served as an Associate Editor for the IEEE TRANSACTIONS ON SIGNAL PROCESSING, an Editor for the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, and an Associate Editor for the IEEE SIGNAL PROCESSING LETTERS. He is a member of the Signal Processing for Communications and Networking Technical Committee of IEEE Signal Processing Society, and a Distinguished Speaker of the IEEE Vehicular Technology Society.



Abbas Jamalipour (Fellow, IEEE) received the Ph.D. degree in electrical engineering from Nagoya University, Nagoya, Japan in 1996.

He holds the positions of Professor of Ubiquitous Mobile Networking with The University of Sydney, Sydney, NSW, Australia. He has authored nine technical books, 11 book chapters, over 550 technical papers, and five patents, all in the area of wireless communications and networking.

Prof. Jamalipour is a recipient of the number of prestigious awards, such as the 2019 IEEE ComSoc

Distinguished Technical Achievement Award in Green Communications, the 2016 IEEE ComSoc Distinguished Technical Achievement Award in Communications Switching and Routing, the 2010 IEEE ComSoc Harold Sobol Award, the 2006 IEEE ComSoc Best Tutorial Paper Award, as well as over 15 Best Paper Awards. He was the President of the IEEE Vehicular Technology Society from 2020 to 2021. Previously, he held the positions of the Executive Vice President and the Editor-in-Chief of VTS Mobile World and has been an Elected Member of the Board of Governors of the IEEE Vehicular Technology Society since 2014. He was the Editor-in-Chief of the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, since January 2022, and the IEEE WIRELESS COMMUNICATIONS, the Vice President of Conferences, and a member of Board of Governors of the IEEE Communications Society. He sits on the Editorial Board of the IEEE ACCESS and several other journals and is a member of Advisory Board of IEEE INTERNET OF THINGS JOURNAL. He has been the General Chair or Technical Program Chair for several prestigious conferences, including IEEE ICC, GLOBECOM, WCNC, and PIMRC. He is a Fellow of the Institute of Electrical, Information, and Communication Engineers, and the Institution of Engineers Australia, an ACM Professional Member, and an IEEE Distinguished Speaker.